

Volume III · Chapter 15 — Evaluation, Bias, Fairness & Clinical Safety · Labs

Source: split from med-tech-curriculum-V1.md (V1). From this split onward this chapter is maintained **independently** here. See Volume-III-Chapter-15-context.md for the AI interaction log.

Prerequisites: Ch 10, Ch 14.

Learning outcomes — the student can: design rigorous clinical evaluation; measure subgroup bias/fairness; assess calibration and safety (red-teaming); explain the benchmark-to-clinic gap; run clinician-in-the-loop evaluation.

Labs (hands-on) — 10 h:

#	Lab	h
15a	Subgroup fairness analysis on a clinical model	3
15b	Calibration curves & reliability diagrams	2
15c	Red-team a medical LLM; catalog failure modes	3
15d	Design a human-eval rubric & run a mini study	2

Datasets/tools: evaluate, fairness libraries; local LLM. **Assessment:** safety/fairness eval report (**60%**); red-team catalog (**20%**); quiz (**20%**). **Key decisions:** which subgroups & metrics; acceptable risk thresholds; when a model is **not** safe to deploy. **References:** plan §13 → *Medical LLMs; Fine-tuning, alignment & benchmarks*. **Hours:** Theory **12** + Lab **10** = **22**.